

Question number	Answer	Additional guidance	Mark
4(a)(i)	An answer that includes: the change in the base sequence of {a gene /DNA/genetic material / allele/genome} (1)	do not accept genetic information accept genome accept ref to change in chromosome{ structure / number} accept RNA	(1)

Question number	Answer	Additional guidance	Mark
4(a)(ii)	An answer that includes the following point: mitosis	do not accept cell division do not accept hybrid mitosis/meiosis	(1)

Question number	Answer	Additional guidance	Mark
4(b)	Choose an item. C is the correct answer A is not the correct answer as three statements are correct. B is not the correct answer as three statements are correct. D is not the correct answer as three statements are correct.		(1)

Question number	Answer	Additional guidance	Mark
4(c)	An answer that includes four of the following points: • moves through the cell membrane /enters the nucleus (1) • oestrogen attaches to (specific){ protein/receptor}(to form a complex) /acts as a transcription factor(1) • which binds to promotor region of {DNA/gene} / allows RNA polymerase to bind to promoter region(1) • which causes (expression / activation/transcription) of a gene that stimulates (cell division/mitosis)(1) • production of a protein involved in {cell division / mitosis}(1)	accept oestrogen enters cell by diffusion ignore receptors cell surface membrane ignore proliferation ignore proliferation	(4)

Question number	Answer	Additional guidance	Mark
4(d)	An answer that includes two of the following points: • acetylation leads to change in {ionic charges / structural change} in histone /DNA (1) • DNA becomes {less/loosely}wrapped (around histones) (1) • transcription can occur(1)	accept neutralizes +ve charges on lysine / histone increases spaces between histones accept RNA polymerase can bind ignore genes are exposed ignore increased gene expression	(2)

